

Predicting Purchase Amount Of Customers During Black Friday

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Abstract—This project predicts Black Friday customer purchases using machine learning and the Black Friday Sales Dataset. Employing Random Forest and XGBoost, the study enhances precision through data exploration, preprocessing, and feature engineering. Unique for its innovative ensemble approach, the project contributes novel insights to machine learning applications, demonstrating XGBoost's superior performance.

Keywords—Black Friday, Purchase Prediction, Machine Learning, Random Forest, XGBoost, Hyperparameter Tuning, Feature Engineering, Data Exploration.

I. INTRODUCTION

Black Friday, an annual shopping extravaganza, marks a pivotal moment in the retail landscape, where consumers unleash their purchasing power in pursuit of enticing deals and discounts. This retail phenomenon not only shapes the financial success of businesses but also unveils a treasure trove of insights into consumer behavior. Understanding the intricacies of Black Friday spending patterns is akin to deciphering the DNA of modern commerce, offering businesses a competitive edge in a dynamic market.

The ability to predict purchasing amounts holds transformative potential for businesses. It enables personalized marketing strategies, facilitates inventory management, and enhances the overall shopping experience. As the retail landscape evolves, driven by technological advancements and shifting consumer preferences, the need to harness data-driven insights becomes more critical than ever.

II. OVERVIEW OF THE PROJECT

A. Problem definition

The problem is formally defined as a Supervised Regression task. The input features include various customer attributes such as age, gender, marital status, and more. The output is the continuous label representing the purchase amount made by each customer during Black Friday. The goal is to build a predictive model that generalizes well to new unseen data.

B. Machine learning framework

The problem is framed as a machine learning task where a model is trained on historical data to learn the relationship between customer attributes and their purchase amount is feature-engineered for accurate predictions. The workflow is shown in Figure 1.

III. DATA EXPLORATION

The dataset used Black Friday Dataset [1] available on Kaggle for purchases consisting of 550,069 rows for predicting sales of customers on black Friday.

The dataset summary statistics in Figures 2 and 3 provide key insights into the distribution and characteristics of important features. The dataset has a diverse range of User_IDs and occupations. Marital status is binary (0 or 1),

with most entries having a value of 0 (not married). Product categories have varying degrees of popularity and purchase amounts exhibit considerable variability. These statistics offer a quick understanding of the dataset's central tendencies, aiding in subsequent analysis and modeling.

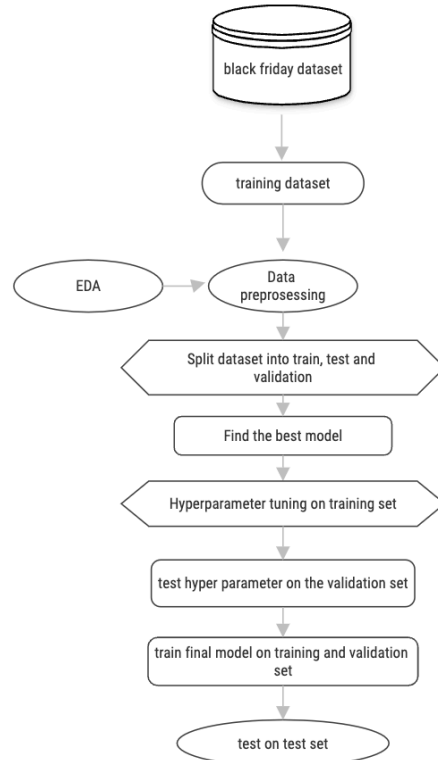


Fig 1. machine learning workflow

	count	mean	std	min	25%	50%	75%	max
User_ID	550068.0	1.003029e+06	1727.591586	1000001.0	1001516.0	1003077.0	1004478.0	1006040.0
Occupation	550068.0	8.076707e+00	6.522660	0.0	2.0	7.0	14.0	20.0
Marital_Status	550068.0	4.096530e-01	0.491770	0.0	0.0	0.0	1.0	1.0
Product_Category_1	550068.0	5.404270e+00	3.936211	1.0	1.0	5.0	8.0	20.0
Product_Category_2	376430.0	9.842329e+00	5.086590	2.0	5.0	9.0	15.0	18.0
Product_Category_3	166821.0	1.266824e+01	4.125338	3.0	9.0	14.0	16.0	18.0
Purchase	550068.0	9.263969e+03	5023.065394	12.0	5823.0	8047.0	12054.0	23961.0

Fig. 2. Statistics Table

‘Product_Category_2’ has 31% empty values which can be replaced by the median, maintaining the original distribution. However, ‘Product_Category_3’ has 69% empty rows so we can drop this feature.

```
User_ID          0.000000
Product_ID       0.000000
Gender           0.000000
Age              0.000000
Occupation       0.000000
City_Category    0.000000
Stay_In_Current_City_Years  0.000000
Marital_Status   0.000000
Product_Category_1  0.000000
Product_Category_2 31.566643
Product_Category_3 69.672659
Purchase         0.000000
dtype: float64
```

Fig 3. Statistics Table

The bar chart in Figure 4 shows the gender distribution, emphasizing a significant male majority. This visual insight prompts consideration of potential gender-related patterns or biases in further analyses.

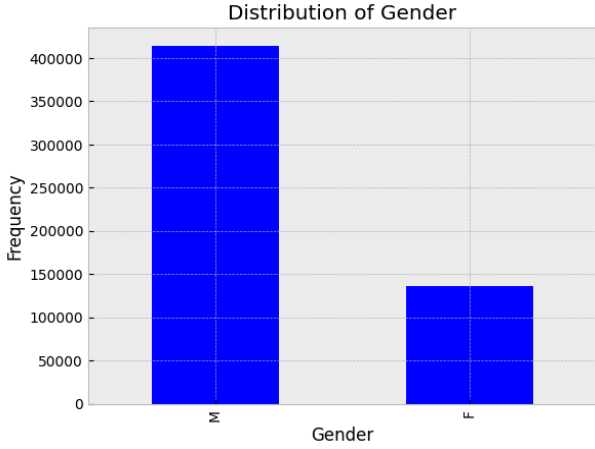


Fig. 4. Purchase difference between men and women

Based on the Black Friday Sales Dataset [1] the heatmap provides us with output as depicted in Figure 5. The observation based on the heatmap is that 'Purchase' is highly related to 'Product_Category_1' and 'Product_Category_2'.

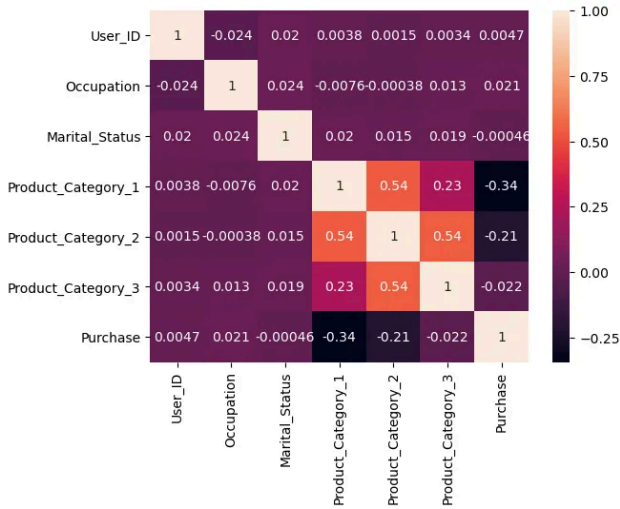


Fig. 5. Heatmap correlation

I then conducted a violin plot analysis to explore the distribution of purchase amounts across different age groups as shown in Figure 6. The purpose was to identify potential outliers or anomalies in the purchase data. The visual inspection showed a consistent purchase pattern across age groups, indicating dataset robustness. The absence of outliers suggests reliable purchase data, minimizing the impact of extreme values on overall patterns.

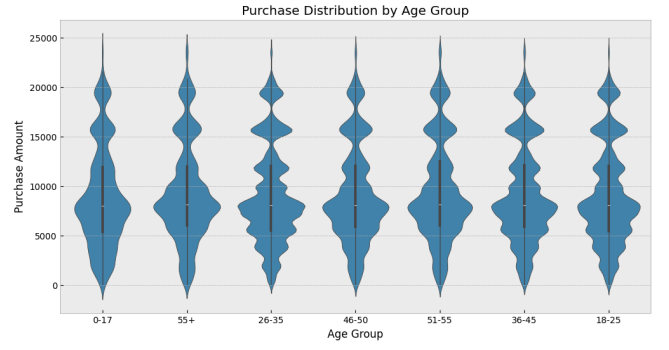


Fig. 6. Violin chart for purchase among ages

The boxplots of numerical columns 'Product_Category_1', 'Product_Category_2', and 'Purchase', offer a visual representation of their distributions and potential outliers as shown in Figure 7. Subsequently, a log transformation was applied to the 'Product_Category_1' column. This transformation serves to mitigate the impact of extreme values, fostering a more symmetrical distribution while preserving the originality of the data. A z-test was also performed to check if the dataset had any other outliers, but none were presented.

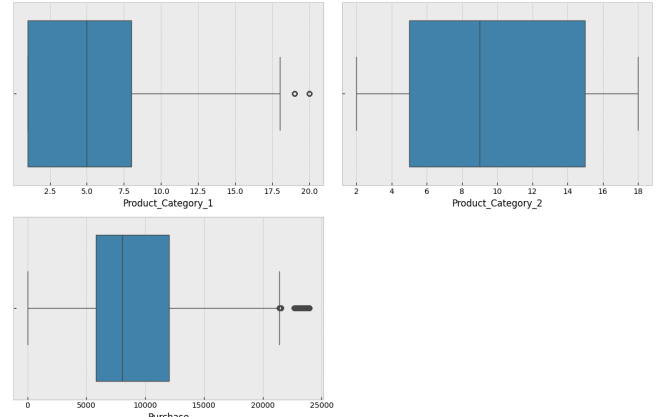


Fig 7. Box plot to check outliers

III. DATA PROCESSING

Performing necessary data preprocessing, we first handle the age and stay duration data features, replacing '55+' and '4+' with '55' and '4', which was noticed by checking for unique values in the column. Subsequently, we remove the 'User_ID' column, which doesn't contribute to predicting the target variable ('Purchase'). Further, we encode the 'Stay_In_Current_City_Years' feature by removing '+' and converting it to numeric format. For encoding categorical features like 'Age', 'City_Category', and 'Gender', we utilize Label Encoder, providing a meaningful representation for model training. The custom mapping for 'Age' ensures an ordered encoding reflecting age groups

IV. DATA MODELING

A. Separating data into test and train

This ensures that you have a training set for model training, a validation set for hyperparameter tuning and model evaluation during training, and a test set for final evaluation after model training and tuning.

B. Feature Importance

Feature selection is performed as shown in Figure 8, before splitting the data into training and testing sets. The reason for this is to avoid data leakage from the test set to the training set, ensuring that the model is evaluated on truly unseen data. This is done to see the importance of columns in the dataset, columns with the least importance are removed to increase future models' accuracy.

```
Gender: 0.47
Age: 2.05
Occupation: 4.87
City_Category: 0.71
Stay_In_Current_City_Years: 2.55
Marital_Status: 0.61
Product_Category_1: 80.97
Product_Category_2: 7.77
```

Fig. 8. Important features

C. Evaluation Matrix

In predicting purchase amounts, Root Mean Squared Error (RMSE) [4] gauges precision, a lower RMSE signifies superior accuracy. R-squared (R2) [4] evaluates the model's capacity to explain purchase variability, ensuring a robust assessment of accurate predictions and understanding underlying purchasing patterns.

D. Model Selection

The choice of Random Forest and XGBoost Regressor [3], rooted in data exploration, aligns with the continuous dataset. Both methods embrace ensemble learning, aggregating weak learners for robustness. XGBoost excels in capturing intricate patterns through sequential tree building, addressing non-linear relationships effectively. Its ability to handle gradients and provide nuanced updates at each boosting round offers a strategic advantage, supported by quantitative measures like lower RMSE as seen in Figure 9, indicating superior predictive accuracy over Random Forest in this task.

After training the model on the training set, we evaluate its performance on the validation set. The model that performs well on the validation set is expected to generalize better to unseen data as shown in Figure 9. This allows us to select the best machine learning algorithm to hyper-tune and get the best results.

E. Comparative analysis

```
Linear Regression Validation RMSE: 4510.156531742059
R2 score: -0.20514952609297632
Random Forest Validation RMSE: 3046.0551562728288
R2 score: -0.6800728651619845
XGB Regressor Validation RMSE: 2969.7038029945356
R2 score: -0.652794757894481
Decision Tree Validation RMSE: 3092.8561746688797
R2 score: -0.6915932942497502
```

Fig. 9. Comparing different machine learning algorithms

V. MODEL EVALUATION

A. Hyperparameter selection

The selection XGBoost Regressor was selected due to its superior handling of non-linear relationships, lower RMSE and r2 than Random Forest, and robust ensemble learning, addressing missing data, feature interactions, and non-linearities for accurate purchase predictions.

The XGBoost Regressor underwent hyperparameter tuning via RandomizedSearchCV, optimizing learning rate, max depth, min child weight, gamma, colsample by tree, subsample, and the number of estimators. The process aimed to minimize mean squared error, employing a five-fold cross-validation strategy for robustness against overfitting, determining selected hyperparameters' impact on model complexity and generalization.

Regularization techniques were applied to strike a balance between adaptability and preventing overfitting. The final model was trained on combined training and validation sets, evaluated on in-training performance through the validation set, and tested for generalization on the test set.

B. Hypertuning the best algorithm

Using RandomizedSearchCV for hyperparameter tuning on the training set and then performing model evaluation on the validation set for an unbiased assessment, comparing the performance of models as shown in Figure 10 compared to Figure 9. Checking for overfitting by comparing the training set with the tuned model validation set in Figure 9.

```
Best Hyperparameters: {'subsample': 1.0, 'n_estimators': 300,
XGB Regressor Tuned Validation RMSE: 2928.6622452660977
```

Fig. 10. RMSE of validation set Hyper-tuned model

A decrease in RMSE indicates an improvement in model performance. In this case, the model seems to generalize better on the validation set after hyperparameter tuning, leading to lower prediction errors.

Finally, train the final model on the combined training and validation sets using the best hyperparameters. By combining the training and validation sets, you make use of a larger labeled dataset for training the final model. This can potentially improve the model's ability to learn and generalize from the available information.

C. Testing on Testing Set

After training, the model is tested on the test set, which has never been seen before. Where the tuned algorithm performs well with decreased RMSE than before, as seen in Figure 11.

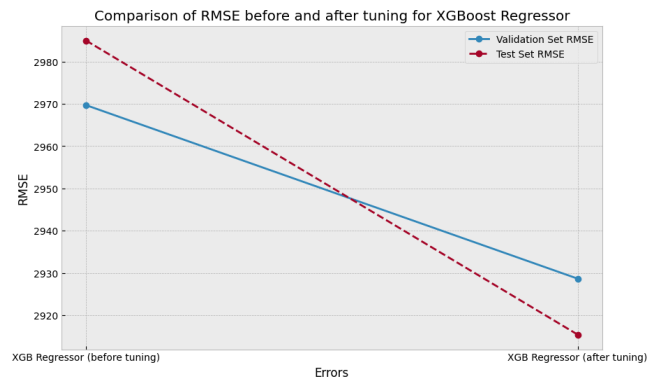


Fig. 11. Tuned XGBoost performance on test set

Line charts in Figure 12 were used to illustrate the trend in RMSE before and after hyperparameter tuning our baseline (Random forest) and proposed model (XGBoost), providing dynamic visualization of the impact of optimization efforts.

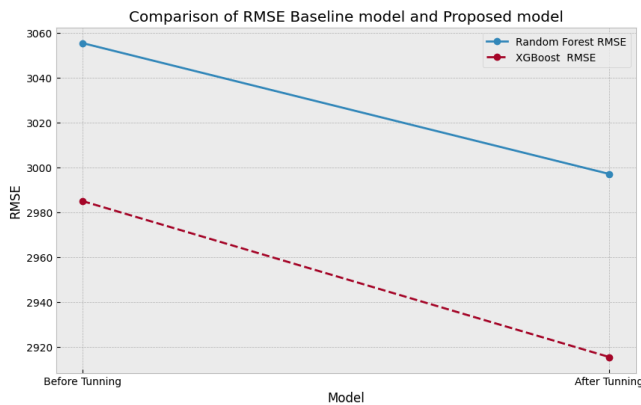


Fig. 12. Comparing Trained Models

VI. SUMMARY OF THE PROJECT

In the course of the project, we conducted data exploration, preprocessing, feature engineering, and feature scaling to prepare the dataset. Our two machine learning algorithms: Random Forest and XGBoost were employed for modeling. Initial results provided baseline and proposed model performance metrics.

Hyperparameter tuning was performed to optimize model performance using the validation set. Then to be tested on the test set. It was observed that XGBoost performed the best with the least RMSE and a higher r^2 score, a decrease in RMSE on the test set highlights a better ability to learn and generalize from the information.

A. Limitations

1. Limited Feature Engineering: The project could benefit from more in-depth feature engineering to capture intricate patterns in customer behavior.
2. Potential biases: various sources, including the data collection process inherent societal biases reflected in the purchase patterns. Unraveling and mitigating these biases demands a delicate balance between thorough exploration, domain knowledge, and a commitment to fairness and equity in predictive modeling.

B. Improvement Strategies

1. Neural Networks:

Artificial Neural Networks (ANNs): Integrate neural networks, particularly suited for capturing complex nonlinear relationships. Explore deep learning architectures like multi-layer perceptrons (MLPs) for regression tasks.

2. Regularization Techniques:

L1 and L2 Regularization: Apply L1 (Lasso) and L2 (Ridge) regularization to prevent overfitting and promote sparsity in the model. Use dropout layers in neural networks to prevent overfitting.

C. lessons Learned

The inherent trade-off between model complexity and generalization becomes apparent, necessitating a careful calibration of hyperparameters to strike the right balance.

This observation underscores the iterative nature of model refinement and the need for thoughtful decision-making in the pursuit of optimal predictive performance.

VII. SELF-EVALUATION

Throughout the lectures, I gained a holistic understanding of the machine learning workflow. Starting with data construction and transformation, I learned to preprocess and engineer features effectively. Exploring a variety of machine learning algorithms, from linear regression to ensemble methods, expanded my knowledge of model selection. The lectures on regularization and cost functions deepened my understanding of preventing overfitting. Additionally, topics like zero-shot learning, classifier evaluation, and clustering provided insights into diverse machine-learning applications. The concepts of dimensionality reduction and hyperparameter tuning equipped me with valuable tools for optimizing models.

Overall, the lectures collectively enhanced my comprehension of the intricacies and practical applications of machine learning algorithms.

The primary difficulty in the module revolved around mastering the intricacies of hyperparameter tuning and ensemble methods, particularly understanding the nuanced impact of different parameters on model performance.

what I took from this coursework was that domain knowledge of the project you do is very important as it can help in feature engineering and selection hence optimizing the model's performance.

If revisiting the predicting purchase amount project, I would intensify feature engineering, explore advanced interpretability techniques with more domain knowledge, and systematically assess diverse machine learning algorithms, including neural networks.

This project's distinctiveness lies in its innovative blend of ensemble learning, encompassing Random Forest and XGBoost, coupled with careful hyperparameter tuning. The tailored feature engineering and insightful data exploration amplify model precision, underscoring its uniqueness. This nuanced combination offers an advanced approach to predicting purchase amounts, contributing novel insights to machine learning applications.

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